$Z\varphi$-classes for $\mathrm{PerfectGroup}(7680, 5)$

v6 orbit–stabilizer / compact-Q computation, $2 \leq n \leq 31$

$n = 16$ computed with the v6.1 abelian-symmetry fast paths

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n	$\lvert S\rvert = 8n$	$\lvert E\rvert = 7680n$	$Z\varphi$-classes	Runtime (s)	Cum. classes	Cum. runtime (s)
2	16	15,360	12	1.466	12	1.466
3	24	23,040	2	1.468	14	2.934
4	32	30,720	350	13.204	364	16.138
5	40	38,400	1	1.410	365	17.548
6	48	46,080	28	13.568	393	31.116
7	56	53,760	1	1.461	394	32.577
8	64	61,440	20,712	215.733	21,106	248.310
9	72	69,120	4	6.533	21,110	254.843
10	80	76,800	24	8.147	21,134	262.990
11	88	84,480	1	1.209	21,135	264.199
12	96	92,160	1,254	116.437	22,389	380.636
13	104	99,840	1	1.571	22,390	382.207
14	112	107,520	24	8.267	22,414	390.474
15	120	115,200	2	9.126	22,416	399.600
16	128	122,880	2,767,210	8,691.149	2,789,626	9,090.749
17	136	130,560	1	4.349	2,789,627	9,095.098
18	144	138,240	72	64.364	2,789,699	9,159.462
19	152	145,920	1	1.307	2,789,700	9,160.769
20	160	153,600	1,244	120.332	2,790,944	9,281.101
21	168	161,280	4	10.976	2,790,948	9,292.077
22	176	168,960	24	9.097	2,790,972	9,301.174
23	184	176,640	1	1.489	2,790,973	9,302.663
24	192	184,320	149,603	2,848.531	2,940,576	12,151.194
25	200	192,000	2	6.759	2,940,578	12,157.953
26	208	199,680	24	8.792	2,940,602	12,166.745
27	216	207,360	12	44.023	2,940,614	12,210.768
28	224	215,040	1,226	108.325	2,941,840	12,319.093
29	232	222,720	1	1.568	2,941,841	12,320.661
30	240	230,400	56	61.692	2,941,897	12,382.353
31	248	238,080	1	1.303	2,941,898	12,383.656

Largest runtime contributions

n	Runtime (s)	Share of total runtime	$Z\varphi$-classes	Share of total classes
16	8,691.149	70.18%	2,767,210	94.06%
24	2,848.531	23.00%	149,603	5.09%
8	215.733	1.74%	20,712	0.70%
20	120.332	0.97%	1,244	0.04%
12	116.437	0.94%	1,254	0.04%
28	108.325	0.87%	1,226	0.04%
	Top six combined	97.71%	2,941,249	99.98%

The complete computation for $2 \leq n \leq 31$ gives **2,941,898** $Z\varphi$-classes in **12,383.656 s** ($\simeq 3.44$ h). The $n = 16$ computation alone gives 2,767,210 classes in 8,691.149 s ($\simeq 2.41$ h), while $n = 24$ gives 149,603 classes in 2,848.531 s ($\simeq 47.48$ min). These runtimes are *not* proportional only to the number of groups S of order $8n$: there are 2,328 groups of order 128 and 1,543 groups of order 192, a ratio $2,328/1,543 \simeq 1.509$, whereas the runtime ratio is $8,691.149/2,848.531 \simeq 3.051$. Equivalently, the average runtime per group is about 3.733 s for $n = 16$ versus 1.846 s for $n = 24$, a factor $\simeq 2.02$. In particular, for $n = 16$, this means that there are exactly **2,767,210 pairwise non-isomorphic extensions** of $\mathrm{PerfectGroup}(7680, 5)$ by groups of order 16.